# Design Review: CARRT Assistive Robotic Arm

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  - Dr. Alqasemi, Leading Professor of CARRT Labs
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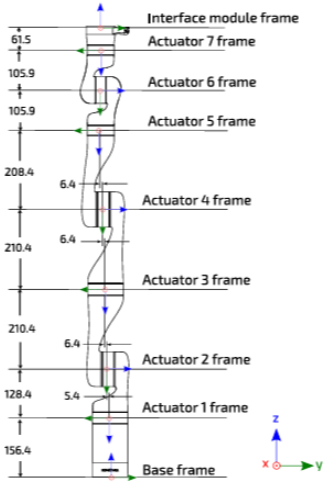
# Agenda

- Background
- Problem
- Requirements
- Design
- Constraints
- Applicable standards
- Risk analysis and mitigation
- Project plan

# Background

- **Approximately 2.5 million people live with moderate to severe mobility impairments of their arm or hands.**
- **The CARRT Lab provided our team with access to the Kinova Gen3 robotic arm.**
- **The ARM is lightweight, programmable and features 7 DoF.**

**KINOVA GEN 3 ROBOTIC ARM**
https://www.kinovarobotics.com/product/gen3-robots




**ARM'S 7 DoF**
https://www.kinovarobotics.com/uploads/User-Guide-Gen3-R07.pdf

# Problem

- **Problem Statement:**
  - Implement three unique ADLs that can be automated by a Kinova Gen3 robotic arm mounted on their wheelchair, specifically for a user with a physical disability affecting upper-limb mobility.
  - Specifically, create a system that can:
    - Clear household objects from table
    - Retrieve a fallen water bottle within range
    - Support medication administration

# Requirements

1. **Accessible Interface:**
   - Touch controls
   - Voice controls
2. **Clear Table:**
   - Identify at least 3 objects on table surface
   - Determine and execute motion based on object
3. **Pick Water Bottle From Floor:**
   - Detect position of water bottle
   - Grasp water bottle and lift to tabletop height
   - Put onto designated area
4. **Medication Administration:**
   - Detect medication bottle using QR code
   - Grasp medication bottle, move it to a user's reachable area
5. **QR-Based Perception**
6. **Operations Run in Simulator First**

# Design – High Level Architecture

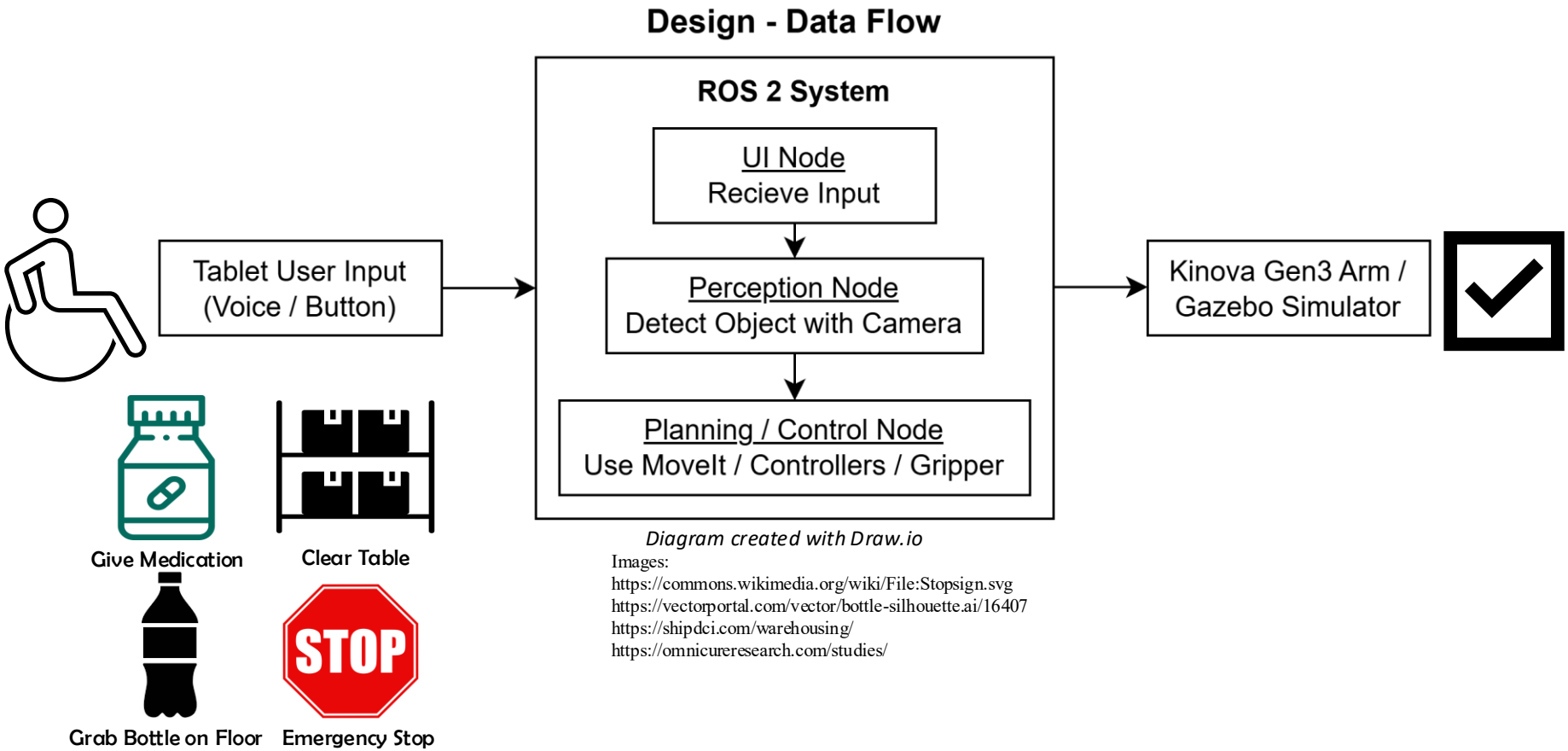
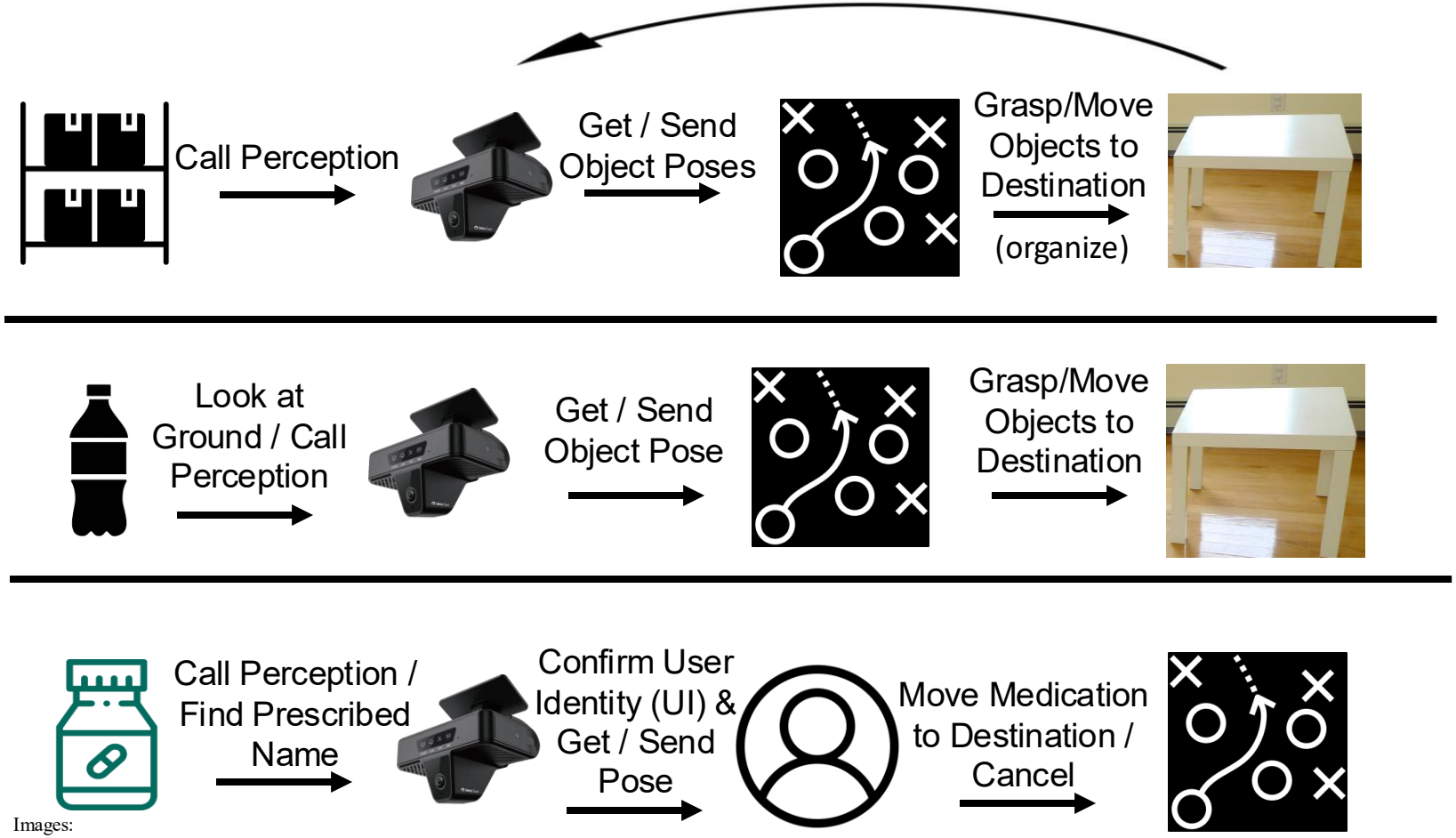



*Diagram created with Draw.io*

Images:
https://commons.wikimedia.org/wiki/File:Stopsign.svg
https://vectorportal.com/vector/bottle-silhouette.ai/16407
https://shipdci.com/warehousing/
https://omnicureresearch.com/studies/

# Design – Data Flow for ADLs




Images:
https://www.safetytrack.com/ai-fleet-dash-cameras/st-220/
https://exitmomentum.com/coaches/ren-flickinger/
https://en.wikipedia.org/wiki/IKEA_Lack
https://freesvg.org/user-icon

# Constraints

- **User**
  - UI must be accessible for persons with disabilities
- **Hardware & Environment**
  - One Arm, Dual Gripper
  - Lab Environment
- **Perception**
  - Simple camera / QR interface
  - Good lighting conditions
- **Object Size**
  - Graspable: 5-15cm width
  - Weight: 1-2kg

Images:
https://freesvg.org/skotan-no-sign-x
https://www.aaastateofplay.com/media/catalog/product/P/l/Plastic_Bowling_Pin_Set_with_5_Pound_Rubber_Bowling_Ball_2.jpg
https://www.flaticon.com/free-icon/approved_7211142
https://www.mahlwerck.de/kaffeebecher-porzellan-bunt-to-go-orange-400-ml

# Applicable standards

- **Robot Safety (Human-Robot Interaction)**
  - ISO 10218-1/2 (2025/2011)
  - ISO/TS 15066 (2016)
    - International Organization for Standardization (ISO)
- **Accessibility / UI**
  - W3C/WCAG Accessibility Guidelines
    - Web Accessibility Initiative
- **Software / ROS2**
  - Naming conventions
  - Simulation standards
  - PEP 8 Python Style

# Risk analysis and mitigation

- **Risks**
  - Software Crash
  - Sensory Misread
  - Logic bugs

- **Mitigation**
  - Auto restart and handle errors.
  - Verify sensor data before acting to avoid mistakes.
  - Test the system thoroughly before real-world use.

# Project plan

- **Key Milestones**
  - Design Review (2/20)
  - Demo Viability Check (3/27)
  - Final Demonstration (5/1)

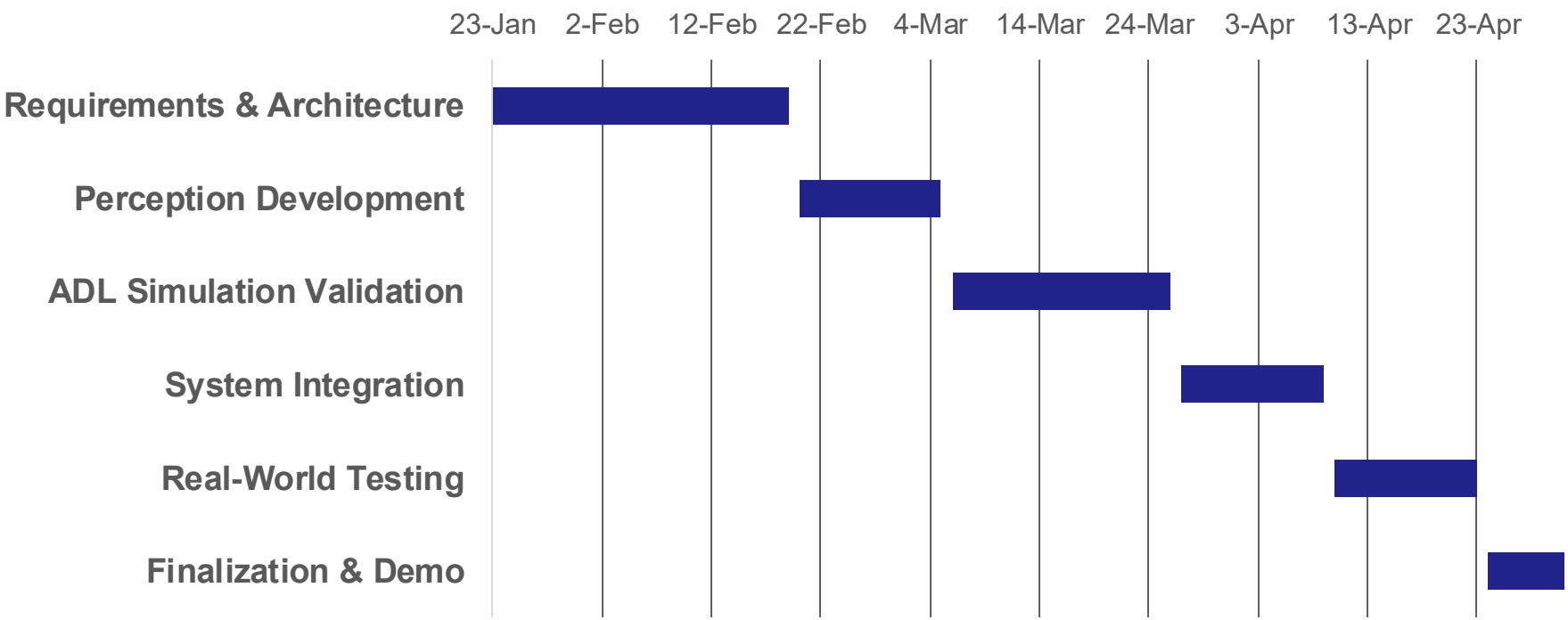



*Diagram created using Excel*